Displaying Time over 4-Digit 7-Segment Display Using Raspberry Pi.

Installation Manual

To display small amount of data with Raspberry Pi, we can use 4 digit 7-segment Display. 7 Segment Display has seven segments in it and each segment has one LED inside it to display the numbers by lighting up the corresponding segments.

Hardware Requirements

1. Raspberry Pi Model A/B/B+
2. 4 digit 7 Segment Display
3. Jumper wires (Female to Female)

Here, I am using **4 digits-7 segments LED** display with TM1637 controller





- **Software Requirements**
  1. Raspbian Stretch OS

1. **Connect your 4 digit 7 segment display with Raspberry Pi's GPIO Pins.**

TM1637 Board Pin	Function	RPI Physical Pin	Raspberry Function
GND	Ground	14	GND
VCC	+ 5V Power	4	5V
DI0	Data In	18	GPIO 24
CLK	Clock	16	GPIO 23

Mr. Prateek Yadav

Code: -

```
import time, datetime, RPi.GPIO as GPIO

CLK, DIO = 23, 24
HEX = [0x3F, 0x06, 0x5B, 0x4F, 0x66, 0x6D, 0x7D, 0x07, 0x7F, 0x6F]

GPIO.setmode(GPIO.BCM)
GPIO.setup([CLK, DIO], GPIO.OUT)

def send_byte(b):
    for i in range(8):
        GPIO.output(CLK, 0)
        GPIO.output(DIO, (b >> i) & 1)
        GPIO.output(CLK, 1)
    GPIO.output(CLK, 0)
    GPIO.setup(DIO, GPIO.IN)
    GPIO.output(CLK, 1); GPIO.output(CLK, 0)
    GPIO.setup(DIO, GPIO.OUT)

def start_stop(func):
    GPIO.output(CLK, 1); GPIO.output(DIO, 1); GPIO.output(DIO, 0); GPIO.output(CLK, 0)
    func()
    GPIO.output(CLK, 0); GPIO.output(DIO, 0); GPIO.output(CLK, 1); GPIO.output(DIO, 1)

while True:
    now = datetime.datetime.now()
    digits = [now.hour // 10, now.hour % 10, now.minute // 10, now.minute % 10]

    start_stop(lambda: send_byte(0x40))  # Write data mode

    def write_digits():
        send_byte(0xC0)  # Start address
        for i, d in enumerate(digits):
            seg = HEX[d] | (0x80 if i == 1 and now.second % 2 == 0 else 0) # Colon blink
            send_byte(seg)

    start_stop(write_digits)
    start_stop(lambda: send_byte(0x8A))  # Display ON, Brightness level 2

    time.sleep(0.5)
```

- **To Run the Program:**
  E.g. Python test.py